ESAKKI RAJAN G

SOFTWARE DEVELOPER | AI & AGENTIC SYSTEMS ENGINEER

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SUMMARY

Software Developer with **5+ years** of experience designing and building scalable web and enterprise SaaS applications using **Python, Node.js, React, TypeScript, SQL, and NoSQL**. Experienced in microservices architecture, **RESTful API development, and cloud deployments on AWS and Azure.** Specialized in **Generative AI, Retrieval-Augmented Generation (RAG), Agentic frameworks, and Human-in-the-Loop (HITL) systems**, with hands-on expertise in LangGraph, prompt engineering, and LLM integration. Strong focus on **performance optimization, secure authentication (JWT, OAuth, RBAC), and message-driven architectures using RabbitMQ.** Proficient in building production-ready AI-powered applications with robust testing (Jest, API Testing) and DevOps automation. Passionate about delivering intelligent, high-performance, and user-centric solutions using modern web and AI technologies.

TECHNICAL SKILLS

- **Programming Languages & Frameworks**: Python, JavaScript, TypeScript, Node.js, React.js
- **AI & GenAI:** RAG Architecture, Agentic Frameworks, LangGraph, Google ADK, Microsoft Agent Framework, Prompt Engineering, HITL Systems
- **Databases & Querying:** PostgreSQL, MongoDB, TypeORM, SQL
- **Cloud Platforms:** AWS (Lambda, S3, CloudFront, API Gateway, Secrets Manager), Azure, Docker, CI/CD (GitHub Actions, Jenkins)
- **Testing & Security:** Jest, Unit Testing, API Testing, JWT, OAuth2, Role-Based Access Control (RBAC)
- **Specialized Skills:** Application Programming, RESTful APIs, WebSockets, Microservices, Message Queues (AWS SQS, RabbitMQ)

PROFESSIONAL EXPERIENCE

Software Developer | LeewayHertz Technologies Pvt. Ltd **Apr 2025 - Present**

- **Developed and deployed** enterprise-grade backend systems using **Python and Node.js**, supporting scalable AI-driven SaaS applications.
- Implemented **Agentic workflows and Generative AI integrations,** enhancing automation capabilities and **improving AI response accuracy by 30%.**
- Designed and orchestrated distributed **microservices on AWS,** ensuring **high availability, performance optimization, and fault tolerance.**
- Integrated **MongoDB and RabbitMQ** for reliable message-driven communication and workflow management across services.

Software Engineer | Ava Software Pvt. Ltd **Mar 2021 - Jan 2025**

- **Developed and deployed** full-stack solutions using **React.js, Node.js, and Python**, enhancing application performance and user experience.
- Implemented **Retrieval-Augmented Generation (RAG) and prompt engineering** for intelligent automation, enabling efficient similarity searches and problem resolution.
- **Designed and optimized SQL queries,** ensuring **data integrity, performance, and scalability** across multiple services.
- Conducted rigorous unit testing using **Jest and API testing tools,** improving system reliability and reducing production bugs.

RECENT PROJECTS

Enterprise AI SaaS Platform

Backend Developer

- Developed and integrated **Agentic AI workflows** into an enterprise SaaS platform using **LangGraph, Google ADK, and Microsoft Agent Framework.**
- Implemented a **Human-in-the-Loop (HITL)** architecture to capture real-time user feedback and continuously improve agent execution quality.
- Built an **Agent-based Retrieval-Augmented Generation (RAG)** system to enhance contextual intelligence, **improving AI response accuracy by 30%.**
- Designed scalable backend services using **Python and Node.js,** enabling reliable AI-driven business workflows.
- Orchestrated distributed AI components using **MongoDB and RabbitMQ** to ensure seamless inter-service communication.
- Deployed and optimized **AI infrastructure on AWS**, ensuring high availability, scalability, and production-grade performance.

AI-Based Service Desk Automation Tool

Full Stack Developer

- Develop an **AI-powered service desk tool,** automating tasks and improving response times.
- Implemented **Retrieval-Augmented Generation (RAG) and Generative AI** to enhance issue resolution using knowledge bases.
- **Optimized latency issues** by designing a **load balancing strategy** across multiple OpenAI models in different Azure regions.
- **Collaborated with cross-functional teams** (HubSpot, IT Security, Cloud Management) to ensure seamless integration and secure operations.

B2C Portal Admin Management System

Node.js Backend Developer

- **Designed and developed** a high-performance API using **Node.js, AWS Lambda, and NoSQL,** reducing response times by **40%.**
- **Enhanced data processing efficiency,** reducing system load and improving response times for large datasets.
- **Collaborated** with cross-functional teams across the **US and Mexico**, ensuring seamless integration and delivery.
- **Implemented security enhancements,** including **OAuth2 and JWT authentication,** improving access control and compliance.

EDUCATION

Bachelor of Engineering (B.E) in Computer Science & Engineering

Anand Institute of Higher Technology, **2021**